

Problem 11.23

At time 0, 100 is deposited into Fund X and also into Fund Y . Fund X accumulates at a force of interest

$$\delta_t = 0.5(1+t)^{-2}.$$

Fund Y accumulates at an annual effective interest rate of i . At the end of 9 years, the accumulated value of Fund X equals the accumulated value of Fund Y . Determine i .

Solution

The accumulated value of Fund X at time 9 is

$$100 \exp \left(\int_0^9 \delta_t dt \right),$$

while the accumulated value of Fund Y is

$$100(1+i)^9.$$

Since the two accumulated values are equal,

$$\begin{aligned} 100(1+i)^9 &= 100 \exp \left(\int_0^9 \frac{0.5}{(1+t)^2} dt \right), \\ (1+i)^9 &= \exp \left(\int_0^9 0.5(1+t)^{-2} dt \right). \end{aligned}$$

Evaluating the integral,

$$\begin{aligned} \int_0^9 0.5(1+t)^{-2} dt &= \left[-\frac{0.5}{1+t} \right]_0^9 \\ &= -\frac{0.5}{10} + 0.5 \\ &= 0.45. \end{aligned}$$

Therefore,

$$\begin{aligned} (1+i)^9 &= e^{0.45}, \\ 1+i &= e^{0.45/9} = e^{0.05}, \\ i &= e^{0.05} - 1 \\ &\approx 0.0512710964. \end{aligned}$$

Thus, the annual effective interest rate is

$$\boxed{i \approx 0.0512711 = 5.1271\%}.$$